

Member 3 Task Sheet

AI Pipeline Developer (Gemini Multimodal OCR, Prompts & Translation)

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1. Core Role Focus

You lead the AI pipeline. Your job is to integrate the Gemini SDK safely, write uploader prompt chains extracting text/metadata from raw notices, configure strict Pydantic JSON schemas, and program English-to-Telugu translation services.

2. Directories to Create and Manage

- **backend/services/** Implement connection clients.
- **backend/routers/** Create API routes mapping upload actions.

3. Target File Checklists & Instructions

A. **backend/services/gemini_client.py** (Day 1 - Highest Priority)

Write wrapper script initializing connection to Gemini. Safely fetch API keys using `os.getenv('GEMINI_API_KEY')` to avoid exposures.

B. **backend/routers/upload.py** (Day 1 - Highest Priority)

Configure uploader endpoints connecting directly to Gemini multimodal models. Feed raw PDFs/images to the API for native layout parsing.

C. **backend/services/analysis_synthesizer.py** (Day 2 - High Priority)

Define structured Pydantic schemas validating LLM outputs. Verify Gemini returns structured fields (summary explanation, deadline lists, law citations) in valid JSON formats.

D. backend/services/chat_synthesizer.py (Day 3 - Medium Priority)

Build prompting parameters merging notice text, chatbot message records, and database legal context for chatbot responses. Force Gemini to ground facts in the RAG laws database, adding mandatory 'informational only' warning tags.

E. backend/services/translator.py (Day 3/4 - Medium Priority)

Write translator endpoints translating summary analyses and checklists to Telugu. Ensure formatting tags and date formats stay unchanged.

4. Daily Schedule Checklist

- **Day 1:** Configure Gemini client connections and write uploader parsing routes.
- **Day 2:** Implement structured JSON validation schemas using Pydantic models.
- **Day 3:** Ground chatbot prompt structures and build translation routes.
- **Day 4:** Test translation accuracy and fine-tune model parameters.